Manoj Mathivanan

Staff Software Engineer | Distributed Systems & Kafka Platforms

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Software architect and technical lead building distributed messaging platforms, cloud infrastructure, and developer frameworks. Experience across PayPal, SAP Labs, and Infosys, with a focus on reliability, automation, and practical AI tooling.

CORE SKILLS

Kafka, Kafka Connect, MirrorMaker 2, Java, Python, GCP, Kubernetes/GKE, Strimzi, Terraform, Helm, Docker, Ansible, Harness CI/CD, mTLS/SASL, Datadog, OpenTelemetry, MCP, REST/OData

EXPERIENCE

PayPal | Feb 2021 - Oct 2026 snapshot

Staff Software Engineer / Kafka Platform

- Architected production Kafka on GKE with Strimzi, Helm, and Harness; automated infrastructure and configuration as code, cutting cluster provisioning from about 3 days to under 15 minutes (2026).
- Redesigned Kafka QA infrastructure using Terraform, Ansible, and Docker; reduced monthly cost from about $39,000 to $4,000 and cluster installation from hours to 5 minutes (2023).
- Designed an autoscaling SNI gateway using one IP per GKE cluster with end-to-end mTLS, highly available discovery, and an architecture-forum-approved cloud migration strategy (2026).
- Built the Kafka MCP server and extended it into Slack troubleshooting; estimated issue data collection and analysis fell from at least 2 hours to under 2 minutes (2025-2026).
- Led MirrorMaker 2 migration from VMs to a Kubernetes-based platform, secure cross-environment replication, SASL adoption, and Datadog observability modernization (2024-2025).
- Contributed Apache Kafka's large-payload performance-test fix (KAFKA-18115, PR #18391), merged upstream in May 2025.
- Built and tested restart-free Kafka cluster switching: producers 1-2 s, consumers 5-10 s, no message loss in three test scenarios; client rollout pending as of October 2026.
- Led architecture reviews, mentoring, and delivery planning for five engineers; covered interim management in 2025 and planned/groomed 351 stories in 2026. Earlier work included regional Kafka setup and automated disaster recovery.

SAP Labs India | Mar 2012 - Feb 2021

Associate Development Architect

- Developed Java frameworks for CDS/OData services, a multi-tenant enterprise backend-as-a-service application, and REST record/playback testing; collaborated with the research team in Germany.
- Automated mobile-platform performance tests from 2 hours to 10 minutes. Built a student-data analysis system for three educational institutes in Uganda and an internal crowdsourcing platform.

Infosys | Jun 2011 - Mar 2012

Quality Engineer

- Tested Citizens Bank's BIC teller application and developed comprehensive test coverage.

SELECTED PROJECTS

Trader: Python market research, backtesting, and paper trading. Kafka bootstrap diagnostics: Java/Spring Boot prototype. IoT: warehouse robot, smart water bottle with reminders and leaderboard, smart helmet, and home automation.

Trader source | Kafka diagnostics source | All projects and demos

EDUCATION

M.Tech, Software Systems - Birla Institute of Technology, Pilani (2017). B.Tech, Computer Science - Amrita University, Coimbatore (2011).

RECOGNITION

3 SAP Catalyst awards (top 5% for three consecutive years); SAP Innovation Week winner (2014, 2016, 2018); SAP d-code Demo Jam finalist (2014); 7 PayPal peer recognitions in 2026.

Career snapshot: October 2026. Staff title documented in 2025. Earlier dates follow the February 2024 resume.